卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-CET-DD-EC-0207 49-B60011S-D0207

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CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

电气 专业 第 02 卷 第 07 册
Electrical SPECIALITY VOLUME BOOK

卷册名称 500kV交流滤波器SC小组接线及布置图
Volume name 500kV SC Filter Single Line Diagram and Layout

批 准 APPROVED BY	邓伟
审 核 REVIEWED BY	田俊强
设计/勘测 DESIGNED BY/ SURVEYED BY	唐华
图 纸 6 张 DRAWING NOS.	打印件 / 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
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3	49-B60011S-D0207-003	500kV交流滤波器SC小组接线及布置图 500kV SC AC Filter Single Line Diagram and Layout Grid Plan	A	1		
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注： 本工程初步设计尚未审定，未通过NTDC审核，图纸存在修改的可能，本册图纸施工前请联系CET确认。

The preliminary design of this project has not been approved and has not been approved by NTDC.

The modification is possible in this diagram. Please contact CET for confirmation before construction.

Description of Construction

1. This volume is the layout and SLD of SC AC filter group. There are 8 SC AC filter groups in the Converter station. For details, see the electrical layout. The rest are the same as this one.

2. The equipment of the SC AC filter group for this project is as follows:

Device name	Supplier
Capacitor C1	Guilin Power Capacitor Co., Ltd.
Reactor L1	Beijing Power Equipment Group Co., Ltd.
Current transformer T1	Hunan Electric Porcelain Appliances Co., Ltd.
Current transformer T2	Anhui Transformer Co., Ltd.
Arrester F1	Fushun Electric Procelain Manufacturing Co., Ltd.
35kV post insulator	Fushun Electric Procelain Manufacturing Co., Ltd.

3. The capacitor incoming line of the SC AC filter group of this project is 2×LGKK-600 wire (listed in the "B60011S-D0203" volume); the connection between the equipment is LJ-300 aluminum stranded wire, and the tube busbar is 6063G-ø100 /90 aluminum tube busbar with damper wire (a ø8 hole should be dug at the bottom of the tubular bus).

4. According to IEC-61936, the clearance at E1 is 4100mm for phase to ground and 5000mm for phase to phase. And according to the calculation report, the clearance at E2 is 900mm for phase to ground and phase to phase; the clearance at E3 is 300mm for phase to ground and phase to phase. The location of E1~E3 are showed in single-line diagram in BA08491S-D0211-002

5. When the equipment foundation is constructed, the electrical installation personnel should closely cooperate, and it should be checked whether the actual equipments at the site are in conformity with the design drawings.

6. Before installing the equipment, carefully read the manufacturer's information and construct according to the installation requirements.

7. The connection between equipment and clap need to follow "1 bolt +2 spring washer +1 nut".

8. The clamps in this volume is listed with reference to the sample of Liaoning Jinxing Electric Power Technology Co., Ltd.

9. SC AC filter group equipment installation see "B60011S-D0208".

卷册说明

1.本卷册用于SC交流滤波器小组内设备平面布置及接线。全站SC交流滤波器小组共有8组,具体布置参见电气总平面;本卷册仅表示其中1组,剩余7组均与其相同。

2.本工程SC交流滤波器小组围栏内设备供货情况如下:

设备名称	供货单位
电容器C1	桂林电力电容器有限责任公司
电抗器L1	北京电力设备总厂有限公司
电流互感器T1	湖南电力电瓷电器有限公司
电流互感器T2	安徽互感器有限公司
避雷器F1	抚顺电瓷制造有限公司
35kV支柱绝缘子	抚顺电瓷制造有限公司

3.本工程SC交流滤波器小组电容器进线采用2×LGKK-600导线（开列在"B60011S-D0203"分册）；设备间连线采用LJ-300铝绞线，管母采用6063G-ø100/90的铝管母线（需在管母最低点加工一个ø8小排水孔），并加装阻尼线。

4.根据IEC-61936，E1点（500kV）对应最小相地距离为4100mm，最小相间距离为5000mm。根据计算结果，电位E2点相间及相地空气净距为900mm；电位E3点相间及相地空气净距为300mm。E1~E3位置参见平面布置图中的配置接线图。

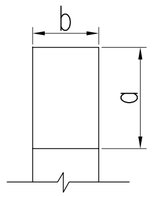
5.土建施工设备基础时，电气安装人员应密切配合，并应核对现场实际到货设备是否与设计图纸相符。

6.设备安装前，应仔细阅读厂家资料并按安装要求施工。

7.交流设备和金具连接应采用"1螺栓+2平垫+1弹垫"。

8.本分册的金具型号参考辽宁锦兴电力金具科技股份有限公司样本中的金具型号开列，设备线夹端子板尺寸示意如下图所示。

9.SC交流滤波器小组设备安装见"B60011S-D0208"分册.



巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程 ±660kV HVDC Project from Matiari to Lahore in Pakistan			NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN
CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.			
拉合尔±660kV换流站 ±660kV Lahore Converter Station		设计阶段 PHASE	施工图 Detailed Design
M L H V D C - L C S - C E T - D D - E C - 0 2 0 7 - 0 0 1 A			
	姓名 Name	日期 Date	卷册说明 Description of Construction
设计 Designed	唐华	2018.12.30	
校核 Checked	曹斌	2018.12.30	
审核 Reviewed	段强	2018.12.30	
批准 Approved		2018.12.30	
比例 Scale	--	图号 DWG NO.	B60011S-D0207-001 版本 REV. A

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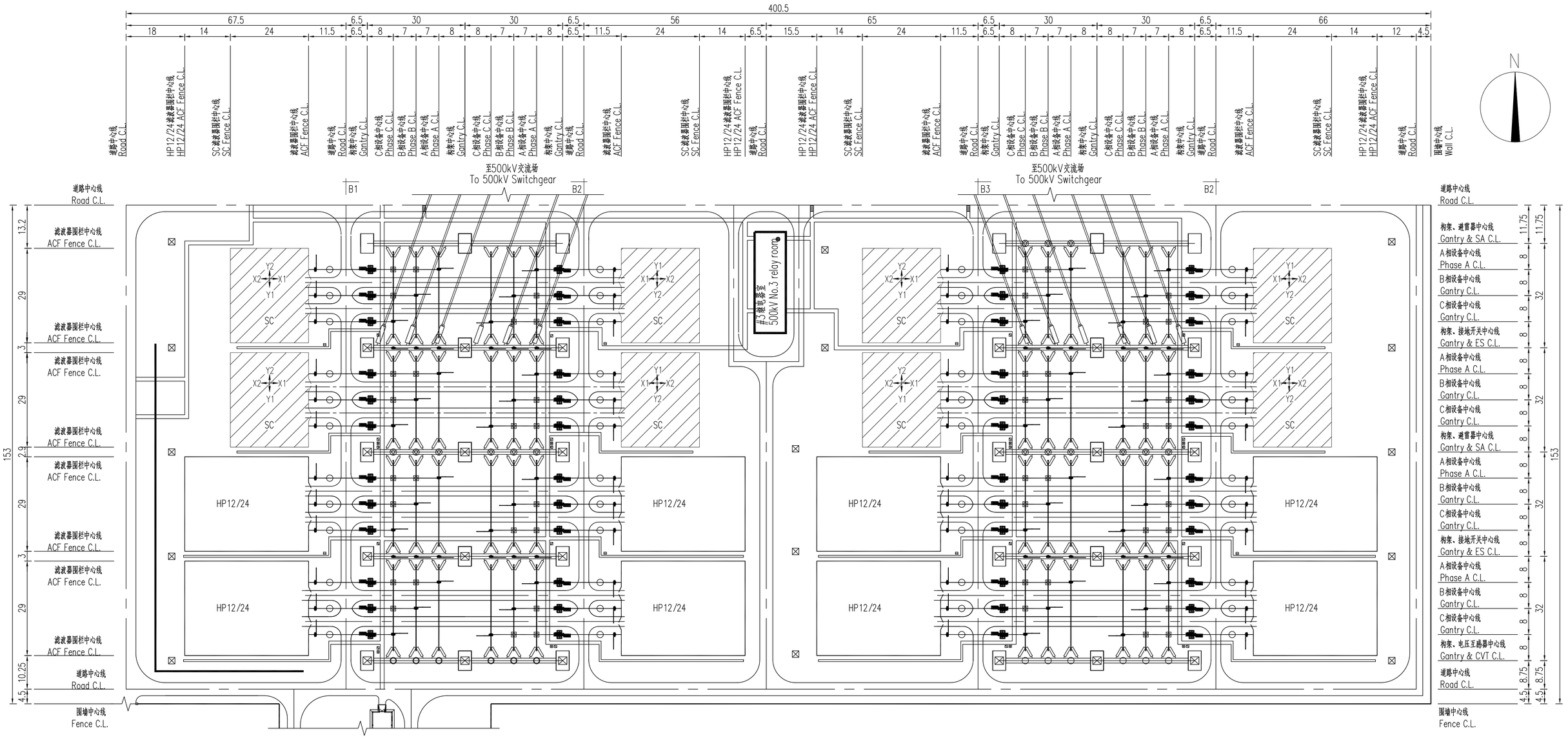


说明：

1. 全站SC交流滤波器小组共有8组。本图为SC交流滤波器的位置图，其相对位置见阴影部分所示。X1、X2、Y1、Y2表示围栏内设备方向。

Note:

1. There are 8 SC AC filter groups in the whole station. This is the location drawing of SC AC filter, and its relative position is shown in the shaded part. X1, X2, Y1 and Y2 represent the direction of equipment inside the fence.

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - C E T - D D - E C - 0 2 0 7 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed		2018.12.30
校核 Checked		2018.12.30
审核 Reviewed		2018.12.30
批准 Approved		2018.12.30
比例 Scale	1:600	图号 DWG NO. 49-B60011S-D0207-002

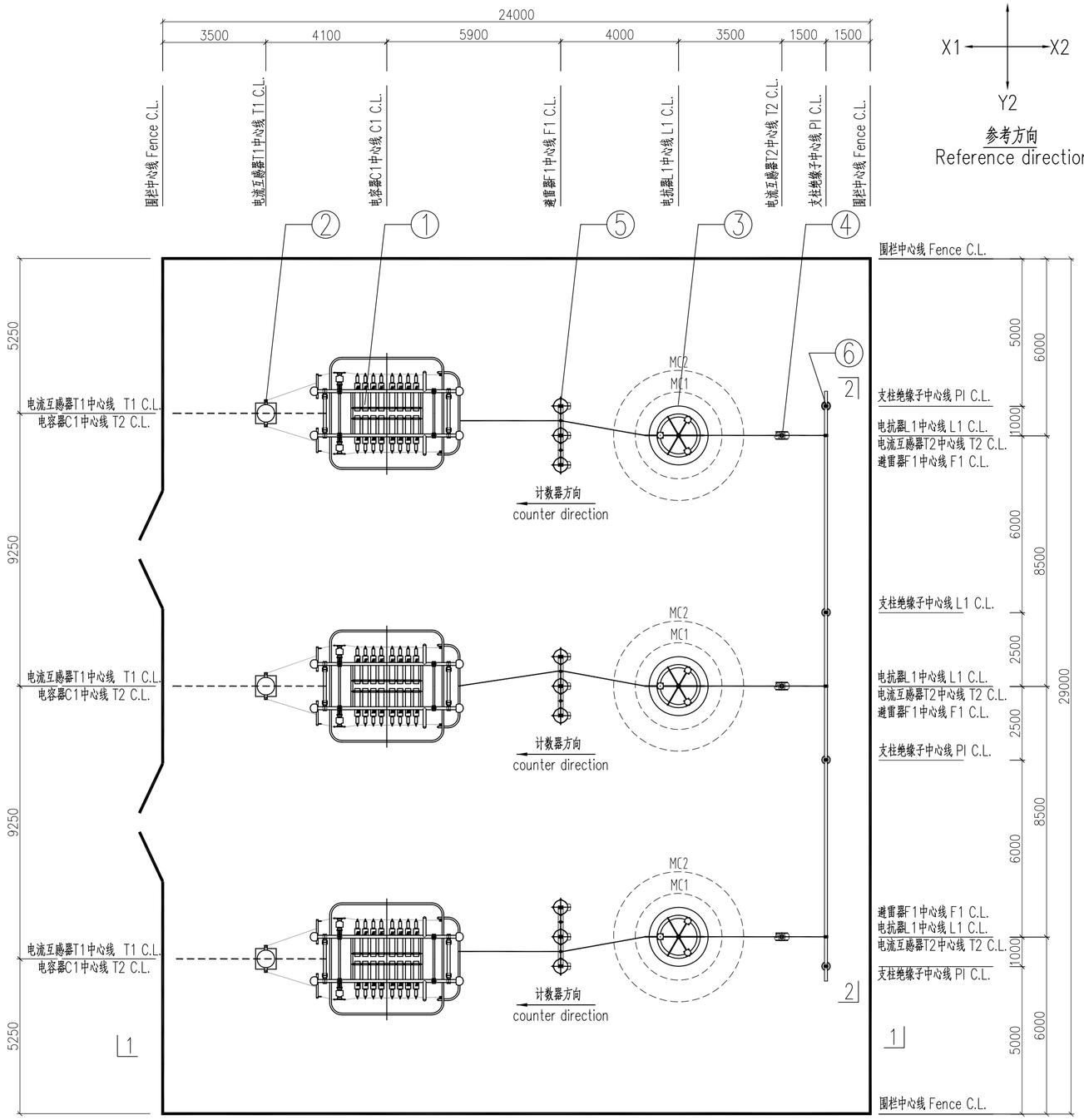
SC交流滤波器小组位置示意图

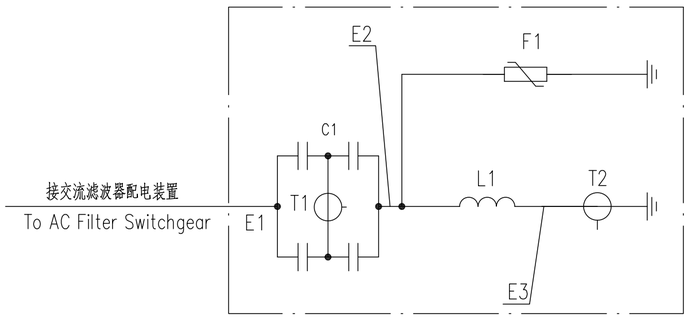
SC AC Filter Group Location Diagram

版本 REV. A

专业 Speciality	会签 Sign	日期 Date
Civil		2018.12.30

594X841





SC交流滤波器小组配置接线图

Single Line Diagram of SC AC Filter

材料表

Table of Equipment and materials

序号	名称	型号及规格	单位	数量 每相	数量 每组	备注
1	电容器C1	TBB510-111552/581-1QW	台	1	3	桂林电力电容器有限责任公司
2	电流互感器(T1)	LB7-330	台	1	3	湖南电力电瓷电器有限公司
3	电抗器(L1)	ZKK-20-290.1-1	台	1	3	北京电力设备总厂有限公司
4	电流互感器(T2)	LZZBJ-24W 500/1A,5P40/5P40/0.2/0.2,5VA	台	1	3	安徽互感器有限公司
5	避雷器(F1)	Y67W1-144/351W	台	1	3	抚顺电瓷制造有限公司
6	支柱绝缘子(PI)	ZSW1.1-40.5/12.5-4	只		4	抚顺电瓷制造有限公司

NO.	Name	Type	Unit	Quantity per phase	Quantity Three phase	Note
1	Capacitor C1	TBB510-111552/581-1QW	Set	1	3	Guilin Power Capacitor Co., Ltd.
2	Current Transformer (T1)	LB7-330	Set	1	3	Hunan Electric Porcelain Appliances Co., Ltd.
3	Reactor (L1)	ZKK-20-290.1-1	Set	1	3	Beijing Power Equipment Group Co., Ltd.
4	Current Transformer (T2)	LZZBJ-24W 500/1A,5P40/5P40/0.2/0.2,5VA	Set	1	3	Anhui Transformer Co., Ltd.
5	Arrester (F1)	Y67W1-144/351W	Set	1	3	Fushun Electric Procelain Manufacturing Co., Ltd.
6	Post Insulator(PI)	ZSW1.1-40.5/12.5-4	Set		4	Fushun Electric Procelain Manufacturing Co., Ltd.

说明:

1. "参考方向"为围栏内设备基础和支架布置方向提供基准,"参考方向"中"X1"指向交流滤波器大组母线侧。全站共8组SC滤波器,请注意各组内设备较围栏相对位置不变,各组间为旋转关系(并非镜像关系)。
2. 避雷器放电计数器朝向如平面布置图中"箭头"朝向所示。
3. 图中尺寸单位为mm。
4. 设备相序与B60011S-D0203图保持一致。

Notes:

1. Referring axis: Y points to main bus of ACF. There are 7 banks of shunt capacitor in total. Note that every bank has the same layout, the position relation of each bank is rotation (not symmetry)
2. The arrows in distribution diagram show the direction of the counter.
3. The unit used in this drawing is mm.
4. The phase sequence of device is consistent with B60011S-D0203.

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - C E T - D D - E C - 0 2 0 7 - 0 0 3 A

	姓名 Name	日期 Date
设计 Designed	唐华	2018.12.30
校核 Checked	高斌	2018.12.30
审核 Reviewed	田俊强	2018.12.30
批准 Approved	刘伟	2018.12.30
比例 Scale	1:120	图号 DWG NO.

500kV交流滤波器SC小组接线及布置图

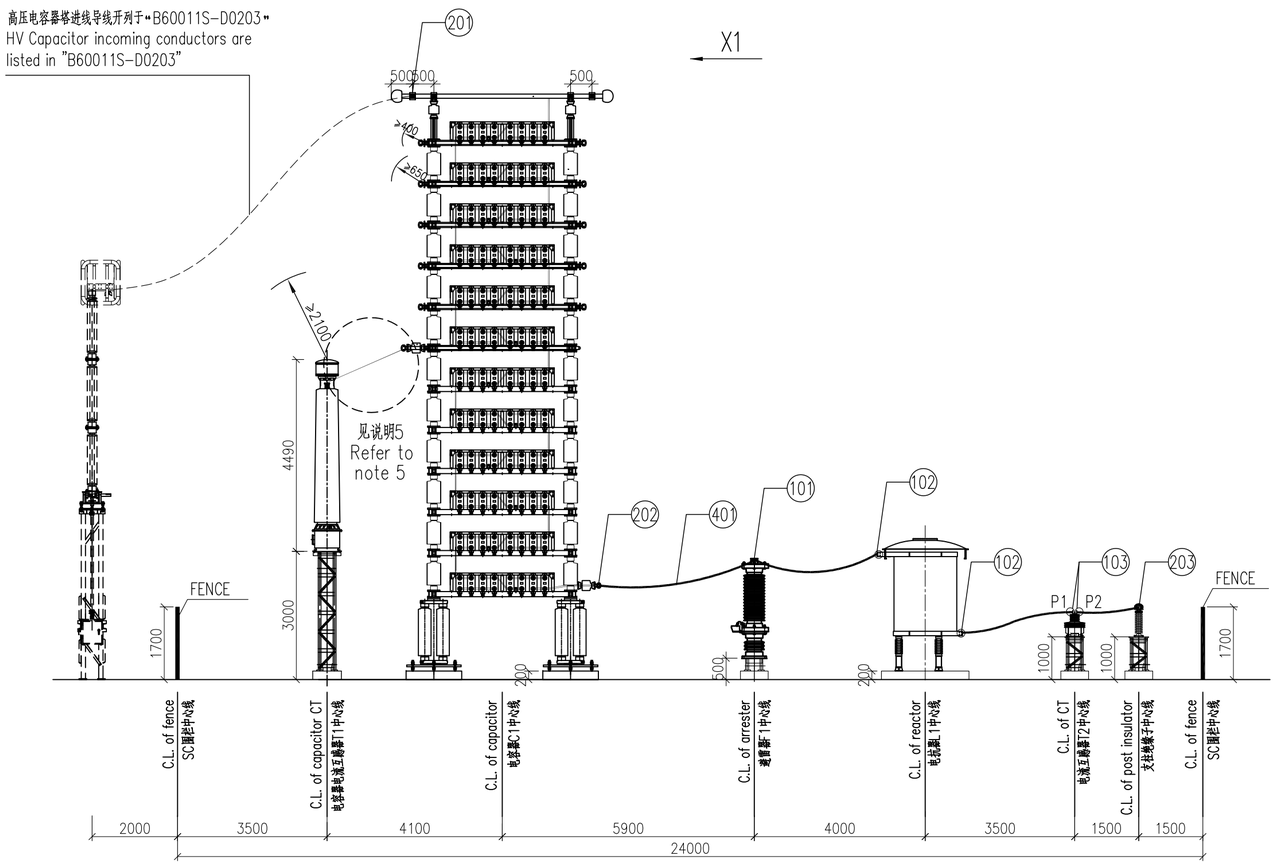
500kV SC Filter Single Line Diagram and Layout

B60011S-D0207-003

版本 REV. A

专业 Speciality	会签 Sign	日期 Date
Civil	张佳	2018.12.30

420X594



1-1断面图

1-1 Sectional view

说明：

1. 在安装设备和导线时，注意满足空气净距要求.
2. 网状遮拦向上延伸线距地2.5m处与遮拦上方带电部分之间的距离按4100mm校验.
3. 高压电容器塔进线采用双分裂导线2xLGKK-600，高压连接金具在本分册开列，导线材料开列在"B60011S-D0203"分册中.
4. 材料表中设备线夹应与铝绞线JL-300-37配套。
5. 此部分连接金具及导线由厂家成套供应。

Note:

1.When installing equipment and conductor, pay attention to the air clearance requirements.
2.The distance between the net-shaped obstruction extending the line 2.5m away from the ground and the live part above the obstruction is verified by 4100mm.
3.The high-voltage capacitor tower conductor Is double-split wire 2xLGKK-600, and the high-voltage clamps are listed in this volume. The conductor materials are listed in the "B60011S-D0203" volume.
4.The equipment clamps in the material table shall be matched with the aluminum stranded wire JL-300-37.
5.This part of clamps and conductor are supplied and installed by manufacturer.

材料表

Table of Equipment and materials

序号	名称	型号及规范	单位	数量 每相	数量 每组	备注	
101	0°设备线夹	SY-300A(165x165)	套	2	6	用于避雷器F1接线	甲供
102	0°设备线夹	SY-300A(102x102)	套	2	6	用于电抗器L1接线	甲供
103	90°铜铝过渡设备线夹	SYG-300C(85x80)	套	2	6	用于电流互感器T2接线	甲供
201	单引流线夹	MGTT-100/LGKK-600Y	套	2	6	用于高压电容器C1高压侧进线 ø100管母连接LGKK-600导线	甲供
202	单引流线夹	MGTT-60/LJ-300Y	套	1	3	用于高压电容器C1低压侧出线 ø60管母连接LJ-300导线	甲供
203	单引流线夹	MGTT-100/LJ-300Y	套	1	3	ø100管母连接LJ-300导线	甲供
401	铝绞线	JL-300-37	米	15	45		

NO.	Name	Type	Unit	Quantity Per phase	Quantity Three phase	Note
101	0° Terminal Connector	SY-300A(165x165)	Set	2	6	For lightning arrester F1 connecting
102	0° Terminal Connector	SY-300A(102x102)	Set	2	6	For reactor L1 connecting
103	90° AL-Cu Terminal Connector	SYG-300A(85x80)	Set	2	6	For current transformer T2 connecting
201	Single jumper clamps	MGTT-100/LGKK-600Y	Set	2	6	For capacitor C1 high voltage side incoming ø100 tube busbar connected to LGKK-600 line
202	Single jumper clamps	MGTT-60/LJ-300Y	Set	1	3	For capacitor C1 low voltage side incoming ø60 tube busbar connected to LJ-300 line
203	Single jumper clamps	MGTT-100/LJ-300Y	Set	1	3	ø100 tube busbars connect LJ-300 line
401	Aluminum conductor	JL-300-37	m	15	45	

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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - C E T - D D - E C - 0 2 0 7 - 0 0 4 A

	姓名 Name	日期 Date
设计 Designed	唐华	2018.12.30
校核 Checked	葛立威	2018.12.30
审核 Reviewed	田俊锷	2018.12.30
批准 Approved	张伟	2018.12.30
比例 Scale	1:100	图号 DWG NO.

SC交流滤波器小组1-1断面图

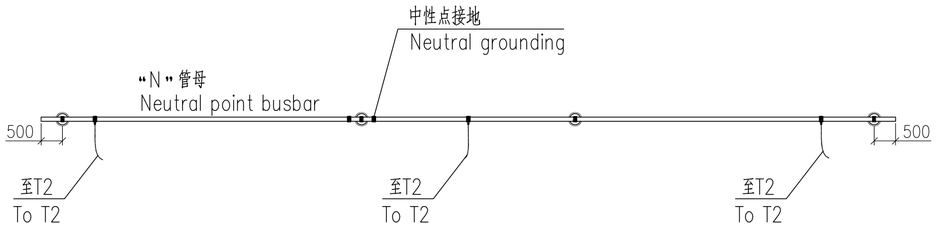
SC AC Filter Group 1-1 Sectional View

B60011S-D0207-004

版本 REV. A

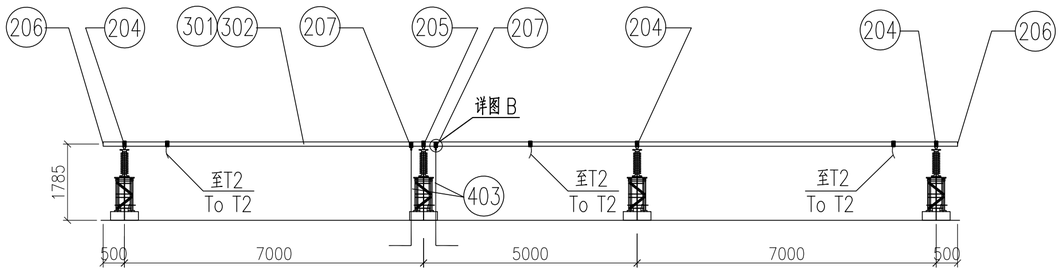
专业 Speciality	会签 Sign	日期 Date
Civil	赵建	2018.12.30

420X594



中性点管母平面布置图

Neutral point tube busbar layout

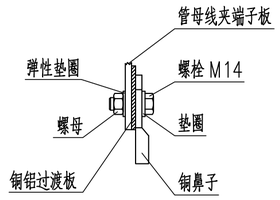


2-2断面图

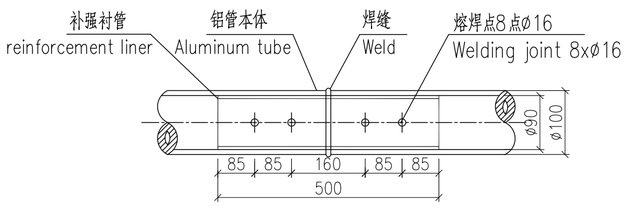
(中性点管母断面图)

2-2 Sectional view

(Neutral point tube busbar sectional view)



详图 B 中性点母线接地引下线接头示意图



详图A

(铝管母线连接图)

Detail A

(Aluminum tube busbar connection diagram)

注: 铝管母线采用氩弧焊接,焊接时,铝管母线接头的两端倒角45°,避免接头外的焊缝与补强衬管焊接在一起,焊好后锉去毛刺.

Note: The aluminum tube busbar is welded by argon arc. When welding, the ends of the aluminum tube busbar joint are chamfered by 45°, Avoid welding the joint with the reinforcement liner, and the burr is removed after welding.

说明:

1. 图中尺寸均以mm为单位.
2. 本断面图表示了SC交流滤波器组管形母线平、断面.
3. 母线由多段管母组装时需增设衬管, 安装方式参见"详图A 铝管母线连接图".
4. 支撑管母设置阻尼线(JL/G1A-300/25钢芯铝绞线), 阻尼线两端固定于管母封端盖上, 阻尼线长度为对应6063G-ø100/90管母长度的110%.

Note:

1. The dimensions in the drawing are in mm.
2. This section shows the layout and section of the SC busbar.
3. When the busbar is assembled by multi-segment tube, it is necessary to add a liner. For the installation method, please refer to "Detail A Aluminium Tube Busbar Connection Diagram".
4. Tube busbar should set the damping wire (ACSR JL/G1A-300/25). Two ends of the damping wire is fixed on the end cap of the busbar, and the length of the damping wire is at least 110% of the length of 6063G-ø100/90 tube.

材料表

Table of Equipment and materials

序号	名称	型号及规范	单位	数量 每相	数量 每组	备注
204	管母滑动线夹	MGG1-100/ø190(4-ø14)	套		3	用于35kV支柱绝缘子 甲供
205	管母固定线夹	MGG2-100/ø190(4-ø14)	套		1	用于35kV支柱绝缘子 甲供
206	管母封堵盖	MGF1-100	套		2	阻尼线型号 JL/G1A-300/25-48/7 甲供
207	管母T接金具	MGT-100A/105x100	套		2	中性点管母接地 甲供
301	铝镁合金管母	6063G-ø100/90	米		20	1根11米 1根9米 甲供
302	铝镁合金管母	6063G-ø89/79, L=500mm	根		3	甲供
402	钢芯铝绞线	JL/G1A-300/25-48/7	米		22	用于ø100/90管母阻尼线 甲供
403	铜绞线	TJ-150	米		10	中性点管母接地 乙供
501	铜铝过渡板	MG-100x10	套		2	乙供
502	放热焊接点	150mm^{2}与120mm^{2}镀铜铜绞线焊接	个		2	用于中性点管母加强接地 乙供
503	铜线鼻	DTM-150	套		2	乙供
NO.	Name	Type	Unit	Quantity Per phase	Quantity Three phase	Note
204	Sliding clamps (Type MGG1)	MGG1-100/ø190(4-ø14)	Set		3	For 35kV post insulators
205	Fixed clamps (Type MGG2)	MGG2-100/ø190(4-ø14)	Set		1	For 35kV post insulators
206	End-balls (with damping clamp)	MGF1-100	Set		2	Damping wire type LJ-300
207	T-connectors (Type MGT)	MGT-100A/105x100	Set		2	Neutral point busbar grounding
301	Aluminum Alloy Tube	6063G-ø100/90	m		20	One is 11 meters long One is 9 meters long
302	Aluminum Alloy Tube	6063G-ø89/79, L=500mm	PCS		3	
402	ACSR	JL/G1A-300/25-48/7	m		22	
403	Copper Stranded Wire	TJ-150	m		10	Neutral point busbar grounding
501	AL-Cu plate	MG-100x10	Set		2	
502	Exothermic welding	A: 150mm² copper strand; B: 120mm² copper strand	Set		2	Neutral point busbar grounding
503	Copper Terminal Lug	DTM-150	Set		2	

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detailed Design

M L H V D C - L C S - C E T - D D - E C - 0 2 0 7 - 0 0 5 A

	姓名 Name	日期 Date
设计 Designed	唐华	2018.12.30
校核 Checked	高成	2018.12.30
审核 Reviewed	田俊强	2018.12.30
批准 Approved	张伟	2018.12.30
比例 Scale	1:100	图号 DWG NO. B60011S-D0207-005

SC交流滤波器小组2-2断面图

SC AC Filter Group 2-2 Sectional View

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420X594

全站SC交流滤波器小组材料表

序号	名　称	型号及规格	单位	数量 每组	数量 全站	备　注
1	电容器C1	TBB510-111552/581-1QW	台	3	24	桂林电力电容器有限责任公司
2	电流互感器(T1)	LB7-330	台	3	24	湖南电力电瓷电器有限公司
3	电抗器(L1)	ZKK-20-290.1-1	台	3	24	北京电力设备总厂有限公司
4	电流互感器(T2)	LZZBJ-24W 500/1A,5P40/5P40/0.2/0.2,5VA	台	3	24	安徽互感器有限公司
5	避雷器(F1)	Y67W1-144/351W	台	3	24	抚顺电瓷制造有限公司
6	支柱绝缘子(PI)	ZSW1.1-40.5/12.5-4	只	4	32	抚顺电瓷制造有限公司
101	0°设备线夹	SY-300A(165x165)	套	6	48	用于避雷器F1接线　甲供
102	0°设备线夹	SY-300A(102x102)	套	6	48	用于电抗器L1接线　甲供
103	90°铜铝过渡设备线夹	SYG-300C(85x80)	套	6	48	用于电流互感器T2接线　甲供
201	单引流线夹	MGTT-100/LGKK-600Y	套	6	48	用于高压电容器C1高压侧进线ø100管母连接LGKK-600导线　甲供
202	单引流线夹	MGTT-60/LJ-300Y	套	3	24	用于高压电容器C1低压侧出线ø60管母连接LJ-300导线　甲供
203	单引流线夹	MGTT-100/LJ-300Y	套	3	24	ø100管母连接LJ-300导线　甲供
204	管母滑动线夹	MGG1-100/ø190(4-ø14)	套	3	24	用于35kV支柱绝缘子　甲供
205	管母固定线夹	MGG2-100/ø190(4-ø14)	套	1	8	用于35kV支柱绝缘子　甲供
206	管母封堵盖	MGF1-100	套	2	16	阻尼线型号JL/G1A-300/25　甲供
207	管母T接金具	MGT-100A/105x100	套	2	16	中性点管母接地　甲供
301	铝镁合金管母	6063G-ø100/90	米	20	160	每根长度为11米，共8根 每根长度为9米，共8根　甲供
302	铝镁合金管母	6063G-ø89/79，L=500mm	根	3	24	甲供
401	铝绞线	LJ-300-37	米	45	360	甲供
402	钢芯铝绞线	JL/G1A-300/25-48/7	米	22	176	用于ø100/90管母阻尼线　甲供
403	铜绞线	TJ-150	米	10	80	中性点管母接地　乙供
501	铜铝过渡板	MG-100x10	套	2	16	乙供
502	放热焊接点	150mm²与120mm²镀铜钢绞线焊接	个	2	16	用于中性点管母加强接地　乙供
503	铜线鼻	DTM-150	套	2	16	乙供

Table of Equipment and materials

NO.	Name	Type	Unit	Quantity per group	Quantity whole Station	Note
1	Capacitor C1	TBB510-111552/581-1QW	Set	3	24	Guilin Power Capacitor Co., Ltd.
2	Current Transformer (T1)	LB7-330	Set	3	24	Hunan Electric Porcelain Appliances Co., Ltd.
3	Reactor (L1)	ZKK-20-290.1-1	Set	3	24	Beijing Power Equipment Group Co., Ltd.
4	Current Transformer (T2)	LZZBJ-24W 500/1A,5P40/5P40/0.2/0.2,5VA	Set	3	24	Anhui Transformer Co., Ltd.
5	Arrester (F1)	Y67W1-144/351W	Set	3	24	Fushun Electric Procelain Manufacturing Co., Ltd.
6	Post Insulator(PI)	ZSW1.1-40.5/12.5-4	Set	4	32	Fushun Electric Procelain Manufacturing Co., Ltd.
101	0° Terminal Connector	SY-300A(165x165)	Set	6	48	For lightning arrester F1 connecting
102	0° Terminal Connector	SY-300A(102x102)	Set	6	48	For reactor L1 connecting
103	90° AL-Cu Terminal Connector	SYG-300C(85x80)	Set	6	48	For current transformer T2 connecting
201	Single jumper clamps	MGTT-100/LGKK-600Y	Set	6	48	For capacitor C1 high voltage side incoming ø100 tube busbar connected to LGKK-600 line
202	Single jumper clamps	MGTT-60/LJ-300Y	Set	3	24	For capacitor C1 low voltage side incoming ø60 tube busbar connected to LJ-300 line
203	Single jumper clamps	MGTT-100/LJ-300Y	Set	3	24	ø100 tube busbars connect LJ-300 line
204	Sliding clamps (Type MGG1)	MGG1-100/ø190(4-ø14)	Set	3	24	For 35kV post insulators
205	Fixed clamps (Type MGG2)	MGG2-100/ø190(4-ø14)	Set	1	8	For 35kV post insulators
206	End-balls (with damping clamp)	MGF1-100	Set	2	16	Damping wire type JL/G1A-300/25
207	T-connectors (Type MGT)	MGT-100A/105x100	Set	2	16	Neutral point busbar grounding
301	Aluminum Alloy Tube	6063G-ø100/90	m	20	160	One is 11 meters long, a total of 8 PCS One is 9 meters long, a total of 8 PCS
302	Aluminum Alloy Tube	6063G-ø89/79, L=500mm	PSC	3	24	
401	Aluminum stranded wire	LJ-300-37	Set	45	360	
402	ACSR	JL/G1A-300/25-48/7	Set	22	176	ø100/90 tube busbar damping wire
403	Copper Stranded Wire	TJ-150	Set	10	80	Neutral point busbar grounding
501	AL-Cu plate	MG-100x10	Set	2	16	
502	Exothermic welding	A: 150mm² copper strand; B: 120mm² copper strand	Set	2	16	Neutral point busbar grounding
503	Copper Terminal Lug	DTM-150	Set	2	16	

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拉合尔±660kV换流站
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M L H V D C - L C S - C E T - D D - E C - 0 2 0 7 - 0 0 6 A

	姓名 Name	日期 Date
设计 Designed	唐华	2018.12.30
校核 Checked	葛立威	2018.12.30
审核 Reviewed	田俊强	2018.12.30
批准 Approved	张伟	2018.12.30
比例 Scale	--	图号 DWG NO.

部分材料汇总表
Partial Material Summary

B60011S-D0207-006

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